

A world map where the landmasses are composed of numerous small, dark-colored coins, likely Euro coins, scattered to form the shapes of continents. The background is a gradient of teal and blue, featuring faint, light-colored circular patterns and numerical markings (150, 160, 170, 180, 220, 230, 240, 250, 260) that suggest a financial or data-driven theme.

CROWDFUNDING

BY

TEAM NEXUS

AGENDA

1. Team Member Roles | Improvements | Responsibilities
2. Problem Statement | Project Description | Team Working Agreement
3. User Personas | MVP | Project Design
4. Technologies | Algorithms | Diagrams
5. Product Backlog
6. Metrics
7. Sprint 2
 - User Stories
 - Test Cases
 - Stories Completed
 - Stories Not Completed
1. Metrics
 - Charts
1. Retrospective
2. Sprint 3
3. Project Demo

GETTING TO KNOW THE TEAM...



Ashwanth Maram
Full Stack Developer



Ashwitha Maram
Developer/Tester



Raj Mandaliya
Block Chain Developer

GETTING TO KNOW THE TEAM...



Vinukonda Yashwanth Reddy
Backend and Database Engineer



Nyi Nyi Lwin
Full Stack Developer



Naveen Pesalabanda
SCRUM Master

IMPROVEMENTS

- Fixed the Diagrams (Architecture Diagram, Context Diagram, ER Diagram, Class Diagram, Sequence Diagram)
- Improve the Metrics (Team Velocity, Sprint Burn down Chart, Completed Committed Ratio)
- Improve user story points and product backlog

PROJECT DESCRIPTION

Project Name:	Crowd Funding
Team:	Team Nexus
Project Description:	It's a platform that helps people raise money safely and fairly using blockchain, making sure all funds are transparent and controlled by the community. For campaign creators and contributors who seek a transparent, secure, and fair way to raise and fund projects, the Crowd Funding platform is a blockchain-based fundraising solution that enables campaign creation, contributor governance, and automatic refunds. Unlike traditional centralized platforms like Kickstarter or GoFundMe, our application uses smart contracts to eliminate intermediaries, reduce fees, and ensure community-driven decision-making.
Benefit Outcomes:	Transparent funds – everyone can see usage Secure money handling – via blockchain Community voting – contributors decide Automatic refunds – if project fails Low fees – cheaper than traditional platforms
GitHub Link:	https://github.com/htmwf2025-Team2/wiki

TEAM WORKING AGREEMENT

Communication Expectations

- Use Zoom, GitHub for collaboration.
- Respect everyone opinions; decisions are made collectively.
- Notifying the team in advance if you cannot complete or attend.
- In case of any conflicts, present research-backed solutions and resolve by majority vote.
- Keeping everyone updated when pushing code or making changes.
- No last-minute dumping of work on others.

Meeting Schedules

- Meetings will be held twice weekly on Zoom or Google meet
- Meeting time is set based on availability.
- Agenda will be shared in advance for preparation.
- Sprint Planning at the start, Retrospective at the end of each sprint.
- Absence must be informed ahead of time; member must read meeting summary.
- All members are expected to participate and contribute fairly.

TEAM WORKING AGREEMENT

Team Dynamics & Principles

- Each member must clearly understand and own their tasks.
- Workload distributed fairly; large tasks broken into smaller ones.
- Every task has a **deadline** and review on completion.
- Encourage problem-solving before escalating blockers.
- Trackers must be updated regularly.
- Work progress should be committed and shared on GitHub.

Checklist for Task Completion

- Code should run **without errors**.
- Work must meet all **acceptance criteria**.
- Each task is **reviewed and tested** by everyone
- Documentation should be updated with the task.
- Final review by all team members before submission.

THE STARTUP FOUNDER PERSONA

Name: Alice Johnson

Age: 28

Gender: Female

Occupation: Startup Founder

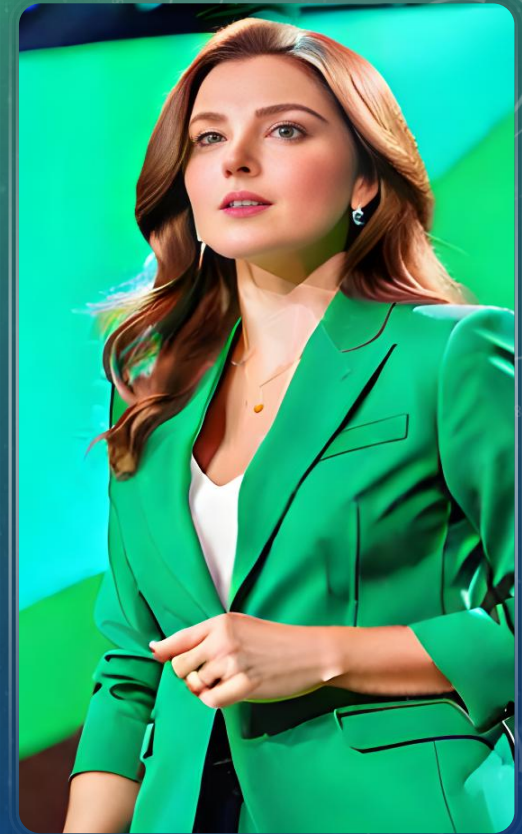
Alice is an ambitious entrepreneur developing a sustainable fashion brand. She wants to raise money to bring her eco-friendly products to market but struggles with high fees and limited visibility on traditional crowdfunding platforms.

Challenges:

- Struggles to gain trust from contributors due to lack of transparency.
- Centralized platforms charge high fees, cutting into her funding.
- Limited control over how her campaign is presented.

Goals:

- Raise funds fairly and transparently.
- Build credibility with backers.
- Ensure contributors see exactly how their money is being used.



THE INVESTOR PERSONA

Name: Bob Smith

Age: 35

Gender: Male

Occupation: Investor

Bob is a passionate investor who likes to fund innovative projects. He wants to support creators but often worries about whether his money is being used properly and feels disconnected after contributing.

Challenges:

- Unsure if funds are spent transparently
- No refunds if a project fails
- Has no control once money is invested

Goals:

- Invest with full transparency and security
- Receive refunds if funding goals aren't met
- Have a say in how project funds are used through voting



THE COMMUNITY MEMBER PERSONA

Name: Cathy Lee
Age: 22
Gender: Female
Occupation: College Student / Community Member

Cathy is an active member of online communities and DAOs. She enjoys contributing to causes but wants her voice to count, not just her money.

Challenges:

- No say in how project funds are allocated
- Hesitant to contribute without accountability
- Finds most platforms profit-driven and restrictive

Goals:

- Participate in voting and decision-making
- Support transparent, community-driven projects
- Learn about blockchain while backing meaningful causes



MVP

- **Campaign Creation:** Allow users to launch funding campaigns with metadata (title, goal, deadline).
- **Contribution Logic:** Accept ETH contributions and track them per user.
- **Refund Mechanism:** Automatically refund contributors if the goal isn't met.
- **Governance Voting:** Let contributors vote on fund allocation or milestones.

TOOLS & TECHNOLOGIES

- **Blockchain:** Ethereum + Solidity
- **Development:** Remix IDE, Hardhat
- **Network:** MetaMask, Sepolia Test Network, Sepolia ETH Faucets
- **Deployment & Testing:** Hardhat
- **Collaboration Tools:** GitHub, VS Code



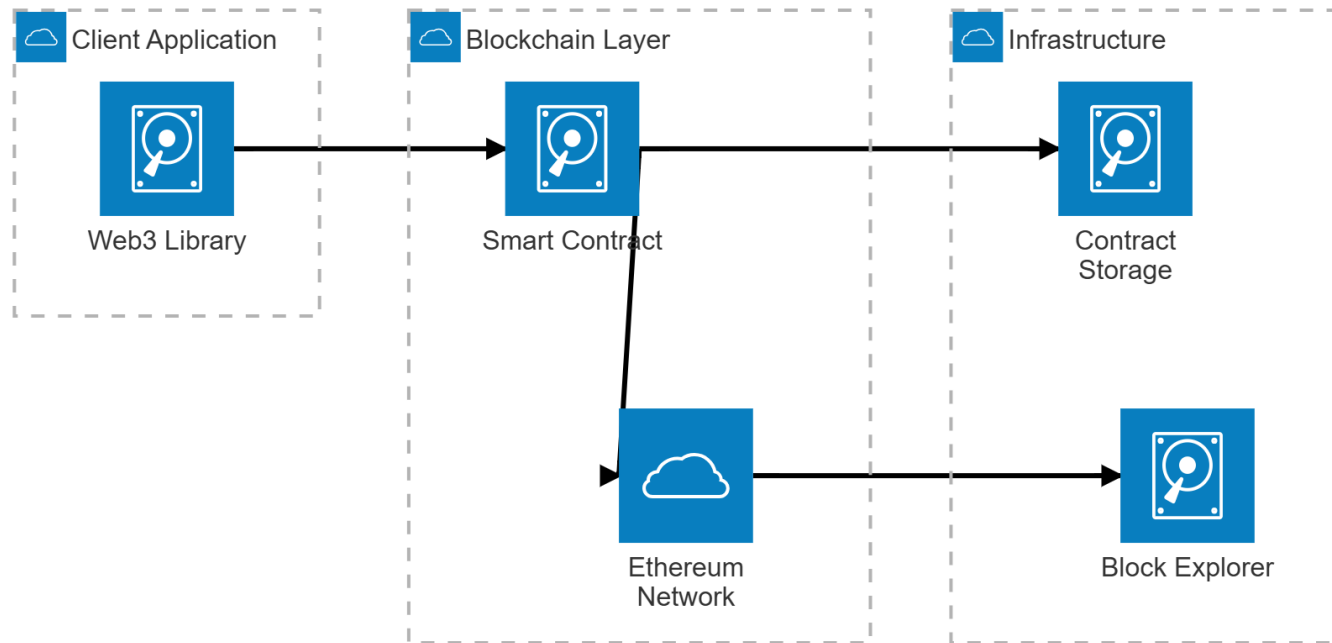
ALGORITHMS & LOGIC

- **Smart Contract–Based Voting Mechanism**
Implements a weighted voting system based on user stake or role.
Ensures tamper-proof, transparent decision-making using smart contracts.
- **Refund Logic**
Automatically refunds users if the funding goal is not achieved.
Executed via smart contract conditions to ensure user trust.
- **Fraud Detection Rules** *(Optional)*
Detects suspicious behavior like duplicate voting or fake contributions.
Helps maintain fairness and integrity in decentralized processes.

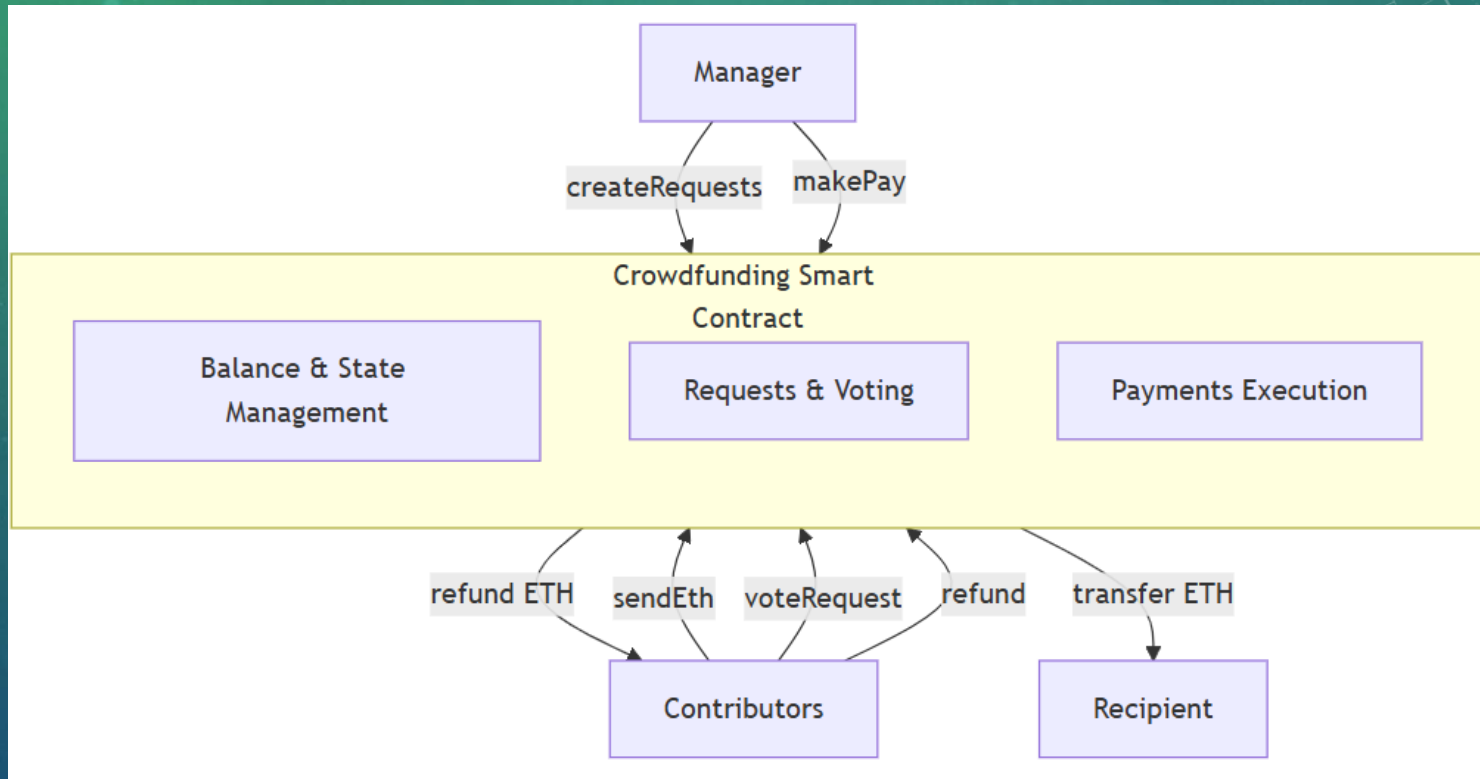
Diagrams

The background features a vertical gradient from green at the top to blue at the bottom, overlaid with a pattern of small white dots. Technical diagrams are visible: a large circular scale with degree markings (0 to 210) and arrows in the top right; a smaller circular diagram with concentric circles and arrows in the bottom right; and partial circular diagrams in the top left and bottom left.

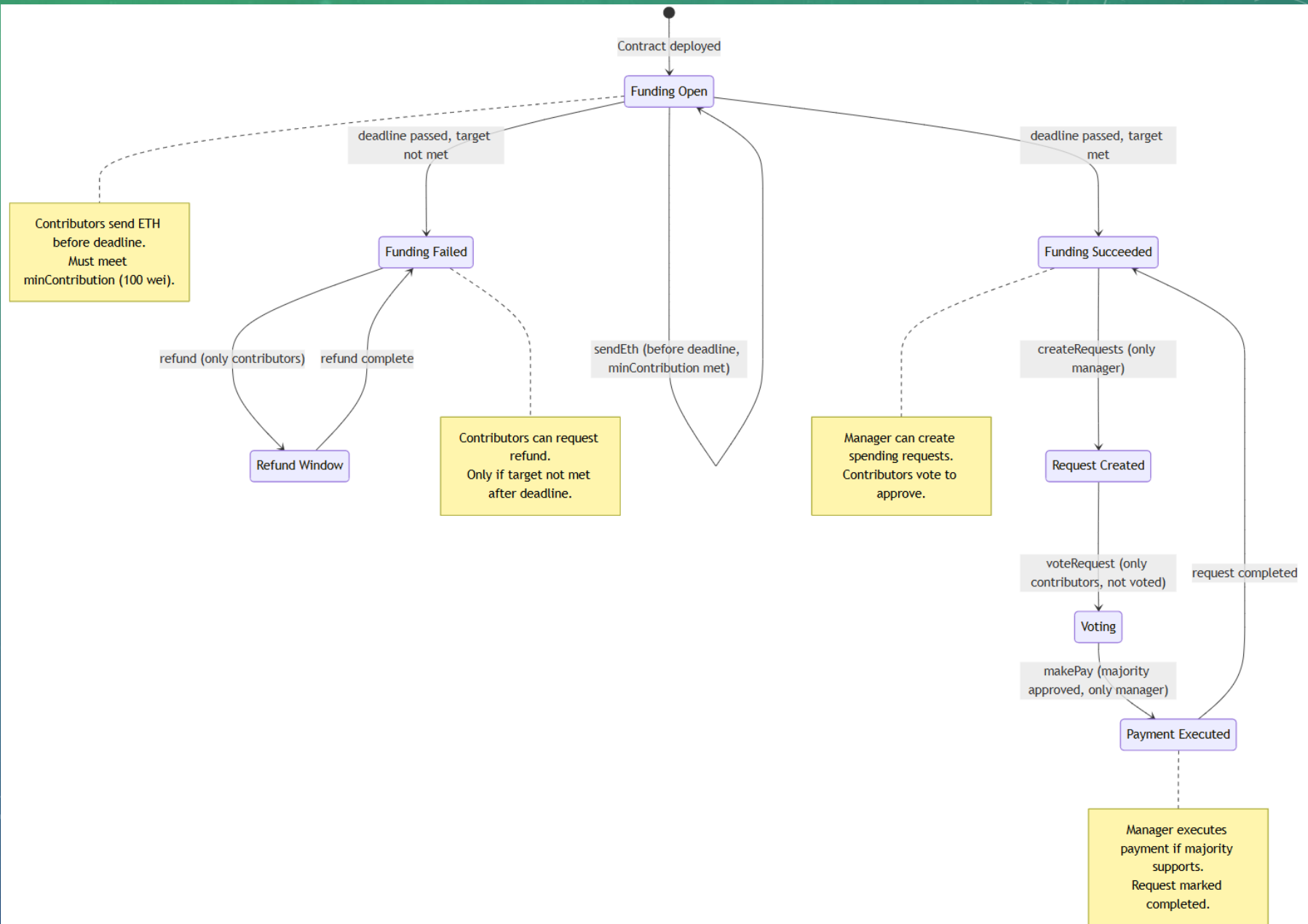
ARCHITECTURE DIAGRAM



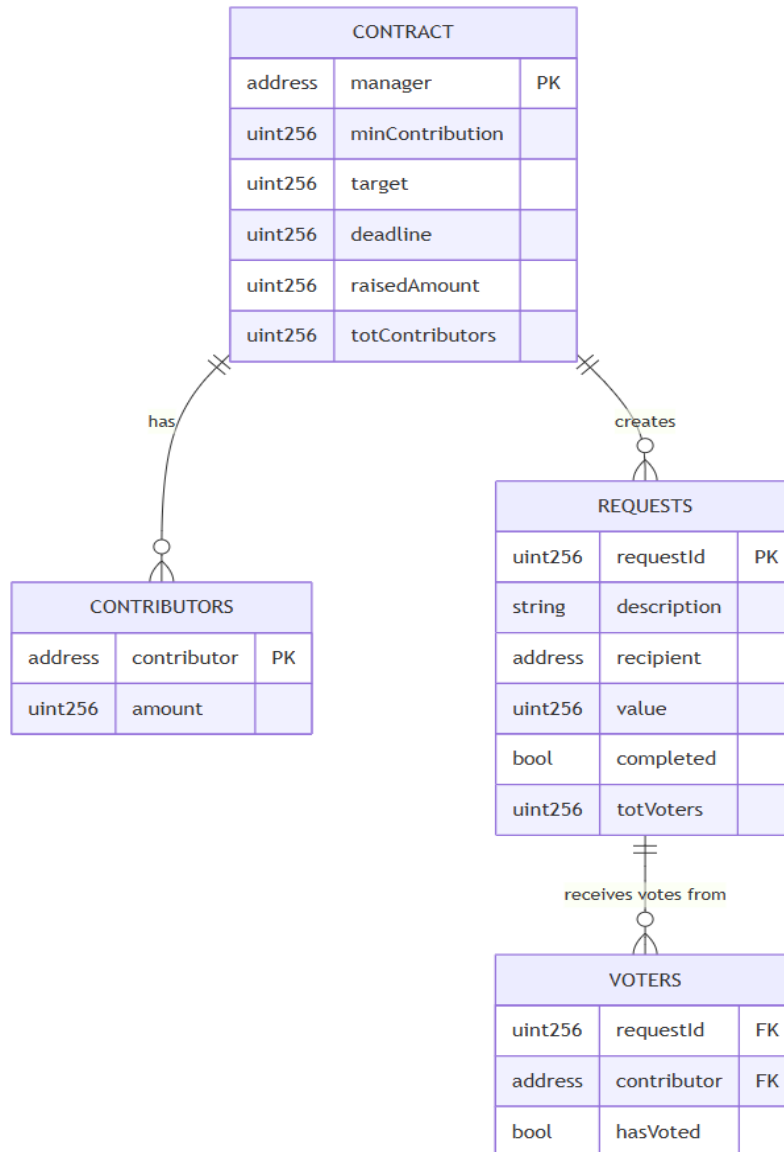
CONTEXT DIAGRAM



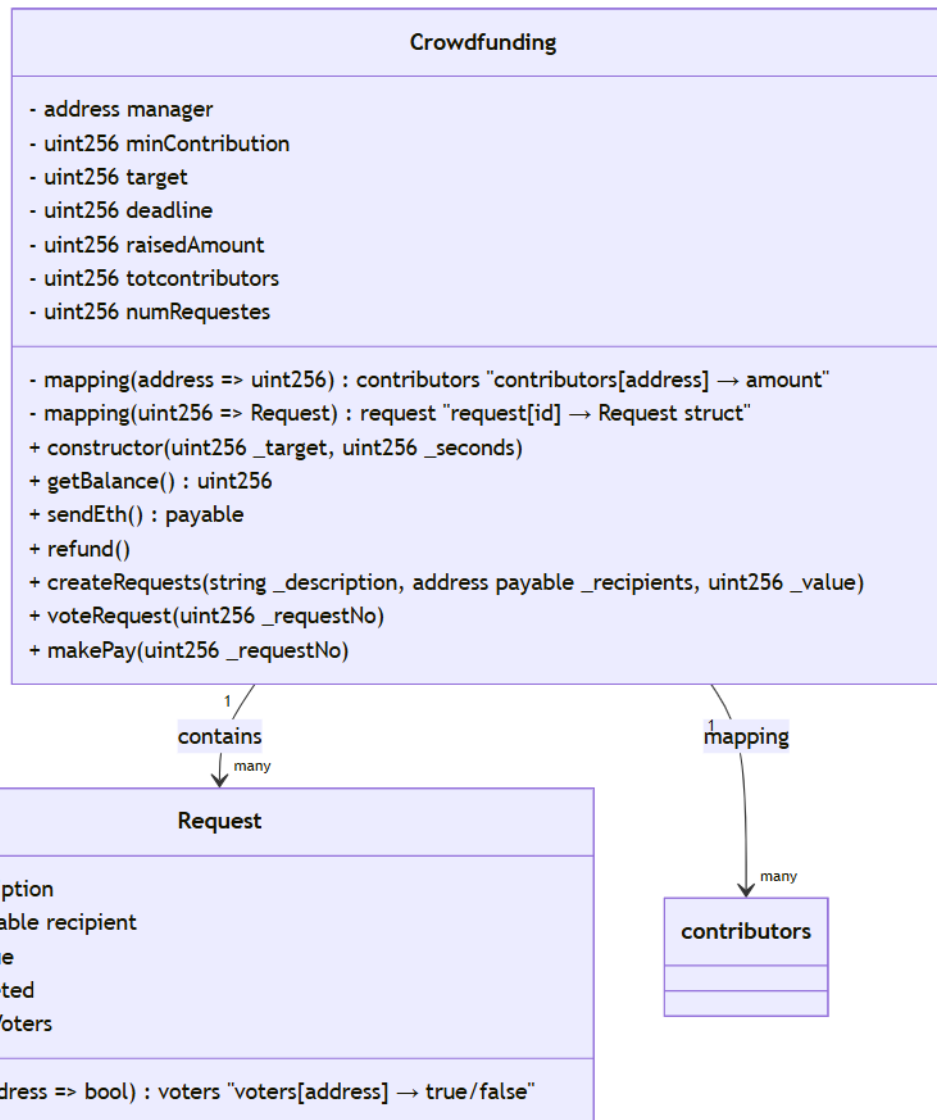
STATE DIAGRAM



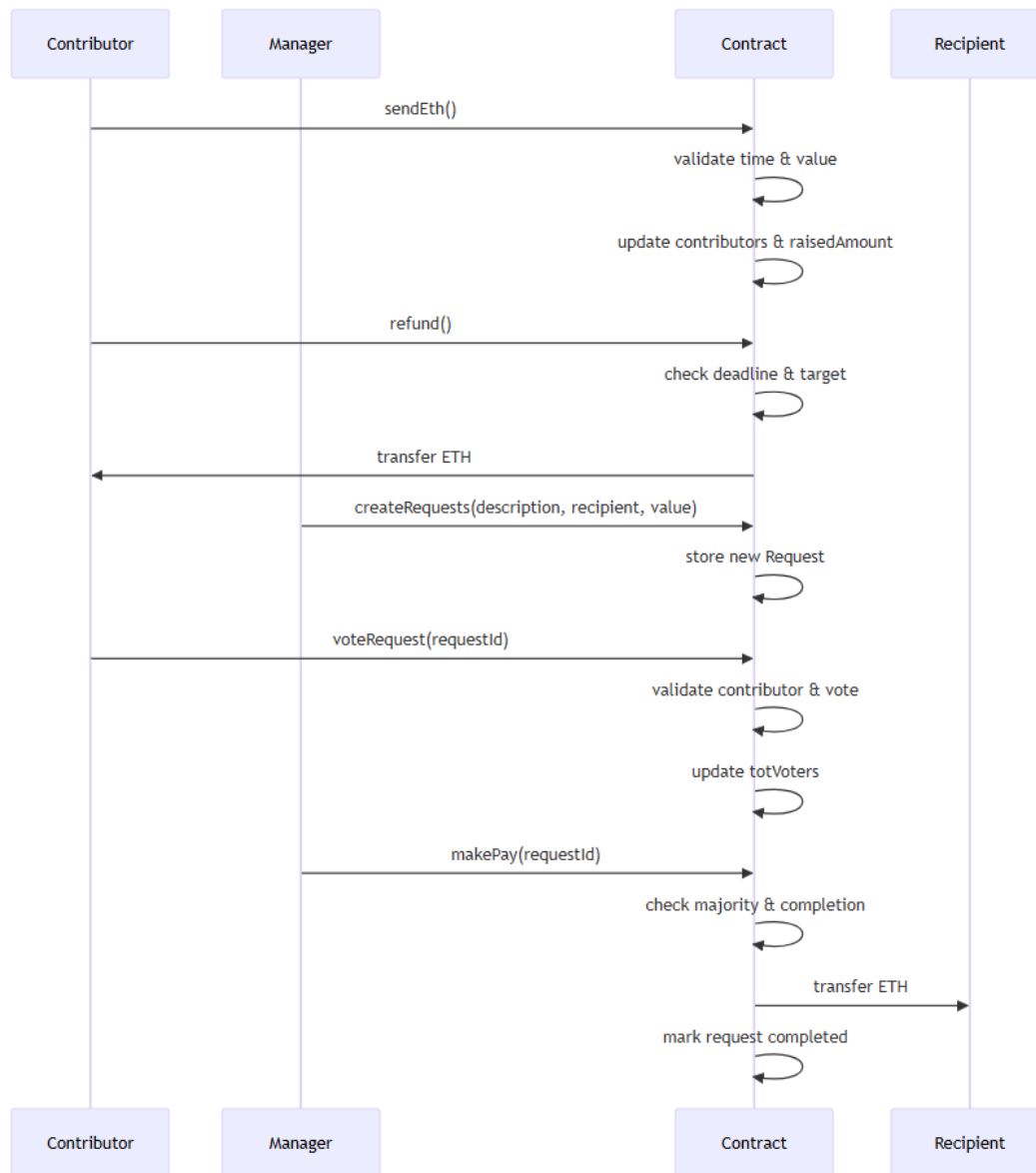
ER DIAGRAM



CLASS DIAGRAM



SEQUENCE DIAGRAM



SPRINT 1 RECAP

PRODUCT BACKLOG



No.	User Story	Acceptance Criteria	Feature Name	Component	Story Points
US_01	As a campaign manager, I want to create a new campaign with a target and deadline so that I can raise funds.	Given I am the manager, When I deploy the contract with a target and deadline, Then the campaign is initialized and I am set as owner.	Campaign Creation	Smart Contract	8
US_02	As a contributor, I want to contribute funds to a campaign so that I can support causes I believe in.	Given the campaign is active, When I send ETH above the minimum, Then my contribution is recorded and the total raised updates.	Campaign Contribution	Smart Contract	5
US_03	As a contributor, I want to get a refund if the campaign fails so that I don't lose my funds.	Given the deadline has passed and the target was not met, When I request a refund, Then I receive my ETH and my record is cleared.	Refund Processing	Smart Contract	5
US_04	As a manager, I want to create spending requests so that I can propose how to use the raised funds.	Given I am the manager, When I create a request with description, amount, and recipient, Then the request is stored with a unique ID.	Request Creation	Governance	8
US_05	As a contributor, I want to vote on spending requests so that I can influence how funds are used.	Given I am a contributor, When I vote on a request I haven't voted on before, Then my vote is recorded and vote count increases.	Request Voting	Governance	5

No.	User Story	Acceptance Criteria	Feature Name	Component	Story Points
US_06	As a manager, I want to release funds after majority approval so that requests can be fulfilled responsibly.	Given the deadline has passed and the target is met, When a request has majority support and I execute it, Then ETH is transferred and request marked completed.	Fund Release	Governance	8
US_07	As a user, I want to view all campaigns with details so that I can choose which to support.	Given campaigns exist, When I open the campaign list, Then I see each campaign's goal, raised amount, and current status.	Campaign Listing	Data Display	5
US_08	As a user, I want to see campaign status so that I know whether it's active, successful, or failed.	Given a campaign has a deadline and target, When time passes or funds are raised, Then the status updates to Active, Successful, or Failed.	Campaign Status	Data Display	3
US_09	As a system, I want to validate minimum contributions so that campaigns remain fair and secure.	Given a contribution is below the minimum, When a user tries to send ETH, Then the transaction is rejected and an error message is shown.	Validation Logic	Contract Validation	2

PRODUCT BACKLOG OVERVIEW

Sprint 1 (15 pts)	Sprint 2 (21 pts)	Sprint 3 (13 pts)
US_01: As a campaign manager, I want to create a new campaign with a target and deadline so that I can raise funds.	US_04: As a manager, I want to create spending requests so that I can propose how to use the raised funds.	US_03: As a contributor, I want to get a refund if the campaign fails so that I don't lose my funds
US_02: As a contributor, I want to contribute funds to a campaign so that I can support causes I believe in	US_05: As a contributor, I want to vote on spending requests so that I can influence how funds are used.	US_07: As a user, I want to view all campaigns with details so that I can choose which to support.
US_09: As a system, I want to validate minimum contributions so that campaigns remain fair and secure.	US_06: As a manager, I want to release funds after majority approval so that requests can be fulfilled responsibly.	US_08: As a user, I want to see campaign status so that I know whether it's active, successful, or failed.

SPRINT 2

SPRINT 2 BACKLOG

No.	User Story	Acceptance Criteria	Feature Name	Component	Story Points	Assigned to
US_04	As a manager, I want to create spending requests so that I can propose how to use the raised funds.	Given I am the manager, When I create a request with description, amount, and recipient, Then the request is stored with a unique ID.	Request Creation	Governance	8	Ashwanth Yashwanth
US_05	As a contributor, I want to vote on spending requests so that I can influence how funds are used.	Given I am a contributor, When I vote on a request I haven't voted on before, Then my vote is recorded and vote count increases.	Request Voting	Governance	5	Nyi Nyi Raj
US_06	As a manager, I want to release funds after majority approval so that requests can be fulfilled responsibly.	Given the deadline has passed and the target is met, When a request has majority support and I execute it, Then ETH is transferred and request marked completed.	Fund Release	Governance	8	Ashwitha Naveen

SPRINT 2 BACKLOG TESTCASES

Feature to Test	Test Case	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
Request Creation	Manager creates request	Call createRequest(description, value, recipient)	desc='Buy Materials', value=2 ETH, recipient=addr1	Request stored with completed=false	Request created successfully, completed=false	Pass
Request Creation	Reject non-manager	Non-manager calls createRequest()	-	Reverts: Only manager can create request	Reverted with correct error message	Pass
Request Creation	Validate fields saved	Create request → check requests[0]	-	Description, value, recipient saved correctly	All values saved and retrieved correctly	Pass
Request Creation	Default state check	Check request.completed	-	completed = false	completed returned as false	Pass
Request Voting	Only contributors vote	Non-contributor calls voteRequest(0)	-	Reverts: You are not a contributor	Reverted with correct error	Pass
Request Voting	Contributor votes once	Contributor calls voteRequest twice	-	Reverts: You have already voted	Second vote reverted as expected	Pass
Request Voting	Vote count updates	3 contributors vote on request 0	3 votes	voteCount = 3	Vote count updated to 3 correctly	Pass
Request Voting	Reject vote on completed request	Finalize → then call voteRequest(0)	-	Reverts: Request already completed	Reverted with correct error message	Pass
Fund Release	Only manager finalizes	Non-manager calls finalizeRequest(0)	-	Reverts: Only manager can finalize	Reverted with correct error	Pass
Fund Release	Majority approval required	Less than 50% votes → finalizeRequest	-	Reverts: Not enough approvals	Reverted with correct error	Pass
Fund Release	ETH transferred	Enough approvals → finalizeRequest(0)	value = 2 ETH	Recipient receives ETH	ETH transferred successfully	Pass
Fund Release	Request marked completed	Call finalizeRequest → check request.completed	-	completed = true	completed set to true	Pass
Fund Release	Prevent double finalization	Call finalizeRequest twice	-	Reverts: Request already completed	Reverted on second attempt	Pass

STORIES COMPLETED

No.	User Stories	Completed Date	Story Points	Completed Points
US_04	As a Manager, I want to create spending requests for raised funds.	Oct 22 - Oct 28	8	8
	Manager creates request.	Oct 22 - Oct 25	2	2
	Reject non-manager.	Oct 25 - Oct 26	4	4
	Validates fields saved & check the default state.	Oct 26 - Oct 28	2	2
US_05	As a contributor, I want to vote on spending requests.	Oct 28 - Nov 6	5	5
	Only contributors vote	Oct 28 - Oct 30	1	1
	Contributor votes only once.	Oct 30 - Nov 3	2	2
	Vote count updates & reject vote on completed requests.	Nov 3 - Nov 6	2	2
US_06	As a manager, I want to release funds after majority approval.	Nov 6 - Nov 12	8	8
	Only manager finalizes & executes payments, majority approval required	Nov 6 - Nov 8	4	4
	ETH transferred & request marked completed & finalization	Nov 8 - Nov 12	4	4

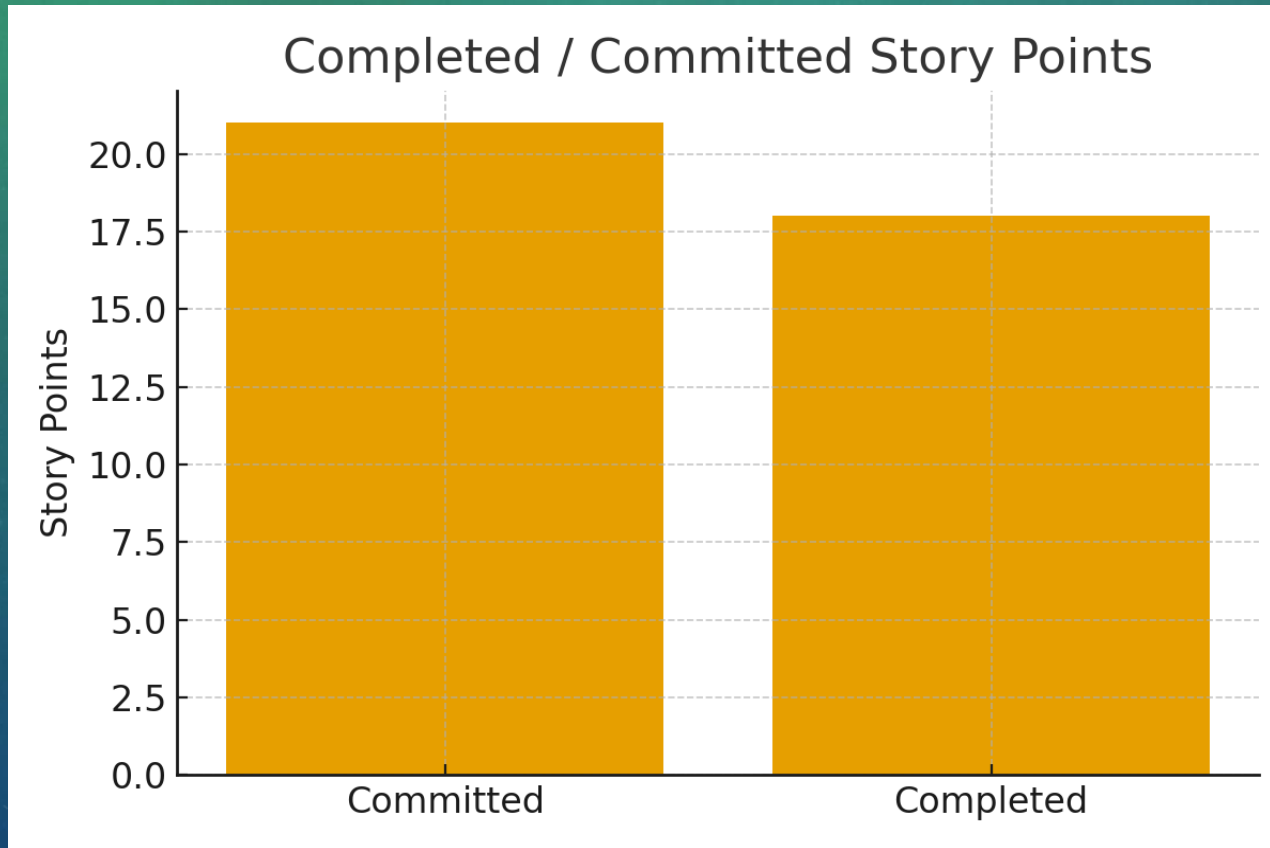
METRICS: SPRINT 2

Completed Committed Ratio:

Current CCR: 1.0, Previous CCR: 1.0, Historical CCR: 1.0

Story ID	Story description	Story points estimated	Story points delivered
US_04	As a Manager, I want to create spending requests for raised funds	8	8
US_05	As a contributor, I want to vote on spending requests.	5	5
US_06	As a manager, I want to release funds after majority approval.	8	8

COMPLETED/COMMITTED RATIO



METRICS: TEAM VELOCITY

Metric	Value
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Total estimated points:	21
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Total points delivered:	21
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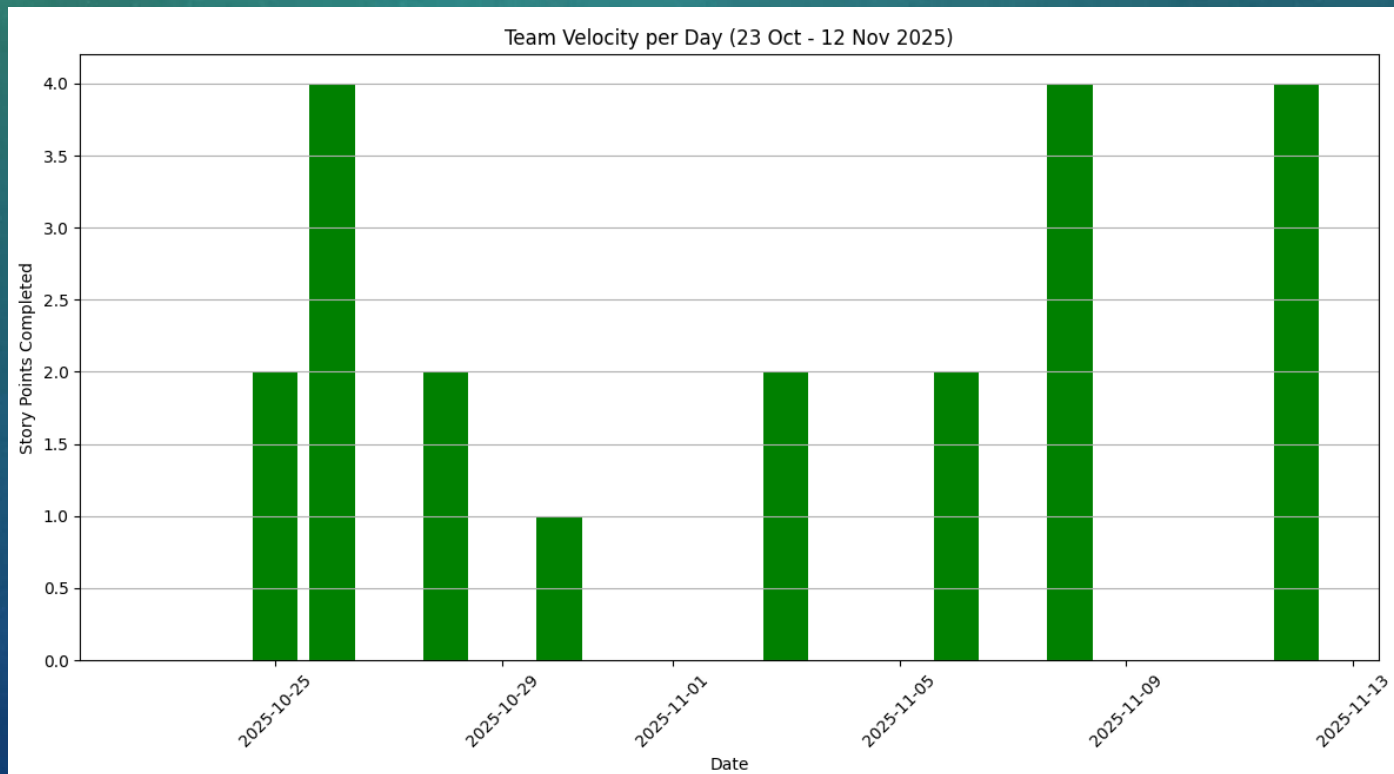
Points carried over:	0
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Team Velocity:	21
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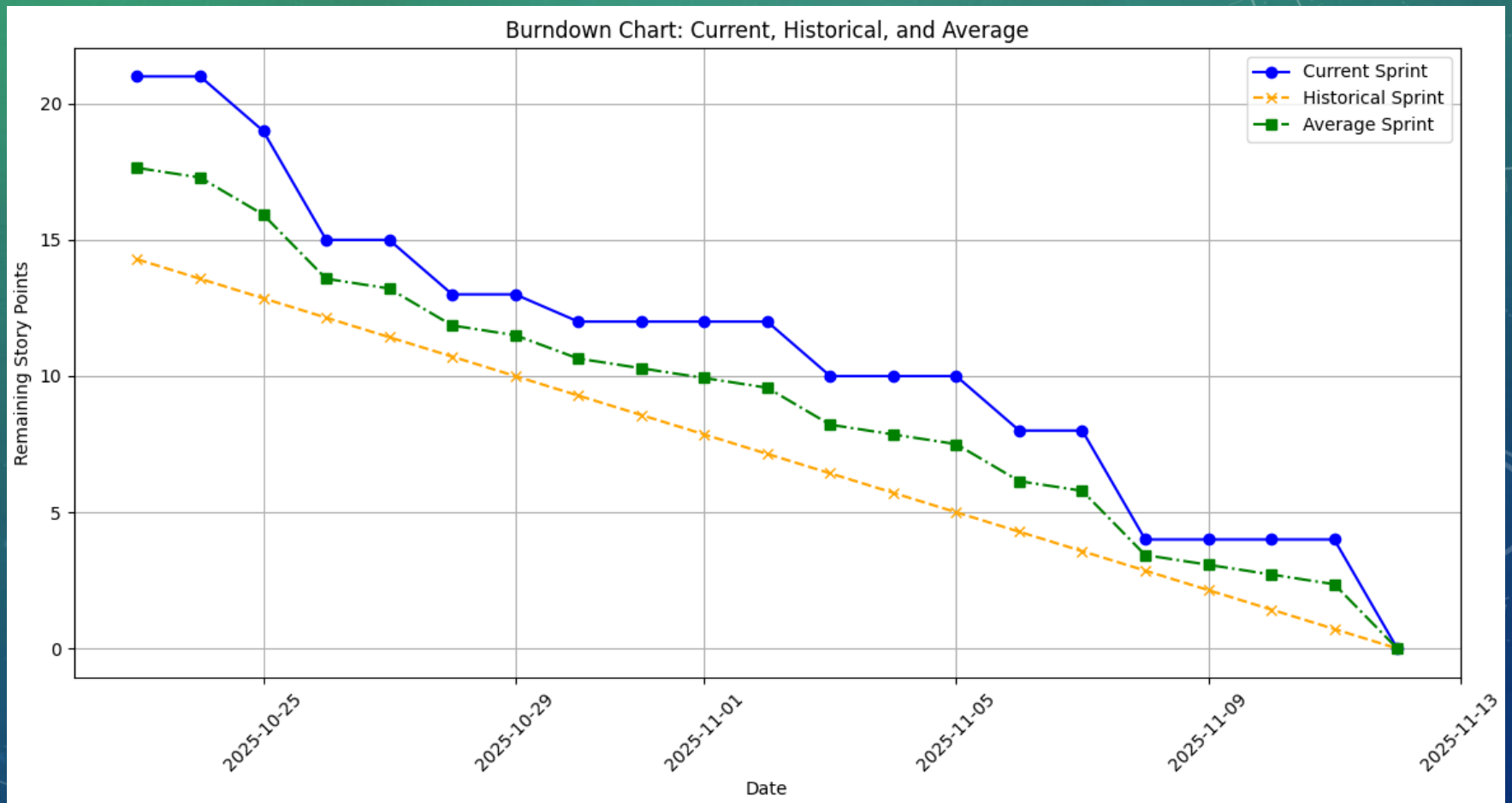
Current Sprint Velocity: 21

Historical Velocity (previous sprint): 15

Average Velocity (all sprints): 18



METRICS: BURNDOWN CHART



SPRINT 2 RETROSPECTIVE

What went well?

- Better coordination among team members.
- Faster task completion compared to Sprint 1
- Clear understanding of the blockchain workflow
- Improved GitHub collaboration and fewer merge conflicts
- Stand-up meetings were more focused and time-efficient
- Test cases were prepared and executed more consistently

What could be improved?

- Need more clarity in dividing complex tasks
- Some stories took longer due to dependency bottlenecks
- Communication gaps during mid-sprint days
- Limited time spent on reviewing pull requests
- Need to improve test coverage and documentation updates

Our commitments

- Break down big stories into smaller, achievable tasks
- Increase communication on blockers in daily stand-ups
- Maintain consistent Jira updates
- Improve unit testing and peer-review process
- Enhance coordination between blockchain and backend integration
- Finish high-priority stories earlier in the sprint

SPRINT 3 PLANNING

No.	User Stories	Acceptance Criteria	Feature	Component	Assigned Task
US_03	As a contributor, I want to get a refund if goal is not met	• Refund only after deadline • Only contributors can request refund • ETH refunded equals original contri-	Refund Processing	Smart Contract	Ashwitha Ashwanth
US_07	As a user, I want to view all campaigns with details	• Show list of all campaigns • Display goal, deadline, raised amount • Update data dynamically	Campaign Listing	Data Display	Nyi Raj
US_08	As a user, I want to see campaign status (Active / Successful / Failed)	• Status depends on raised amount and time	Campaign Status	Data Display	Ashwitha Naveen

SPRINT-3 STORIES PLANNED

No.	User Stories	Acceptance Criteria	Planned Dates	Story Points
US_03	As a contributor, I want to get a refund if the goal is not met before the deadline	• Refund available only after deadline • Only contributors can request refund	Nov 20 to 22	25
US_07	As a user, I want to view all campaigns with details	• Display campaign list • Show goal, deadline, raised amount	Nov 22 to 25	20
US_08	As a user, I want to see campaign status (Active, Successful, Failed)	• Status updates based on raised amount and time • Proper label displayed	Nov 25 to 27	20

PROJECT DEMO



WIKI PAGE LINK

- <https://github.com/htmwf2025-Team2/wiki>



Thank You!